

The Impact of Screen Time

How digital habits affect mental wellness, productivity, & sleep

Jay & Ryan · Data Science Final Project

Why Does Screen Time Matter?

Average daily screen time has more than doubled over the past decade. Its effects on mental health and productivity are still poorly understood. Screens are all around us, but what are the downsides?

6+ Hours

Average daily screen time for U.S. adults in 2024

#1

Mental health cited as top concern among college students

Screen time is rising fast

Smartphones, remote work, and streaming have pushed daily usage to record highs with no sign of slowing down.

Mental health is declining

Rates of anxiety and depression, especially among young adults and students, have risen alongside screen adoption.

Productivity is at stake

Employers and educators report concentration and output issues increasingly linked to digital distraction and overuse.

Leisure is controllable

Unlike work-related screen use, leisure screen time is discretionary, meaning it can actually be reduced.

Research Overview

Research Question: How does screen time affect mental wellness, productivity, stress, and sleep quality?

400

Individuals

15+

Variables

3

Screen Types

Key Variables Tracked

- Screen hours (total, work, and leisure)
- Mental wellness index (0–100)
- Productivity score (0–100)
- Stress level (0–10)
- Sleep hours & quality
- Exercise & social hours
- Occupation & work mode

Source:

Kaggle dataset: *Screen Time vs. Mental Wellness Survey 2025* by Adharshinikumar

-Available at:

kaggle.com/datasets/adharshinikumar/screentime-vs-mentalwellness-survey-2025

Dataset overview:

- 400 survey respondents across a range of ages, occupations, and work modes
- No major cleaning was required, only variable names were standardized for consistency

Methodology

Three analytical approaches were used to answer: **how does screen time affect wellbeing, and who is most at risk?**

01

Correlation Analysis

What: Pearson correlation coefficients between screen time (total and leisure) and each wellbeing variable: mental wellness, productivity, stress, and social hours.

Why: Establishes which outcomes are most strongly linked to screen time before modeling.

02

Stepwise Regression

What: A stepwise selection ($p\text{-entry} = 0.05$) on all non-collinear predictors to identify the strongest drivers of total daily screen time.

Why: Determines which individual characteristics, like occupation, work mode, stress, productivity, predict how much screen time a person accumulates. When we know what is most important, solutions are easier to implement.

03

Linear Regression + Interaction

What: A linear model predicting mental wellness from screen time: a baseline model and an interaction model adding social hours as a moderator.

Why: Quantifies the precise cost of each extra hour of screen time, and tests whether socializing buffers the harm.

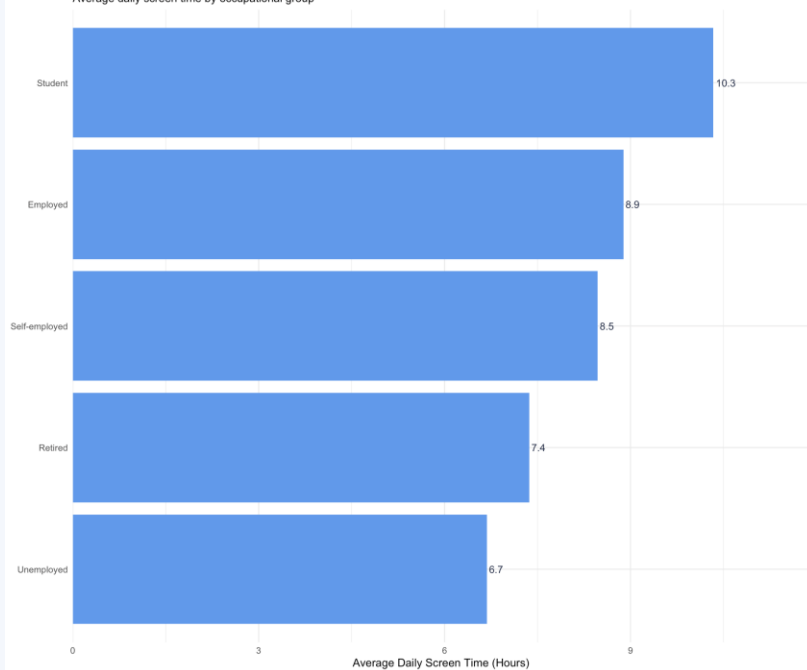
Who Uses Screens the Most?

By Occupation

Key Finding: Students accumulate the most screen time (~10.3 hrs/day), followed by employed individuals. Also, leisure screen time accounts for the majority of total screen time across all occupations

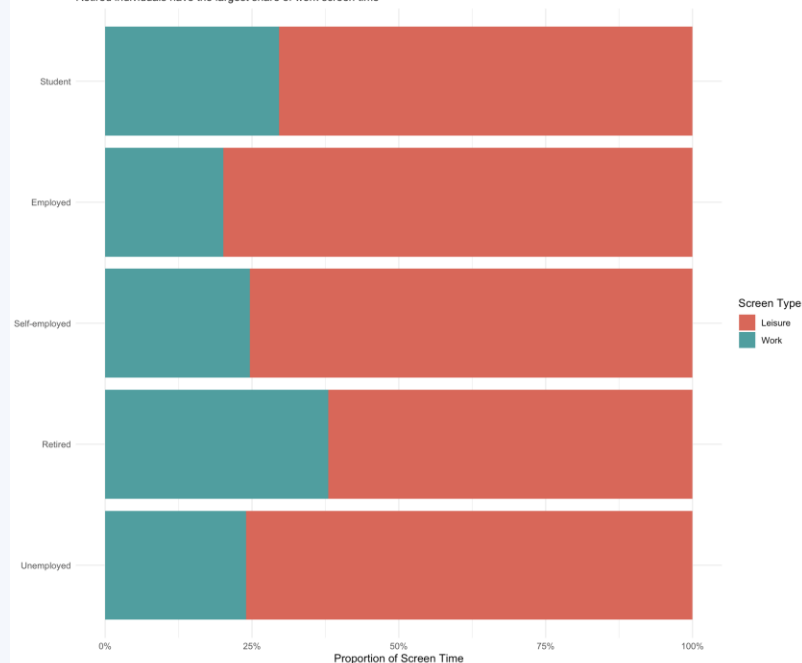
Students Spend the Most Time on Screens

Average daily screen time by occupational group



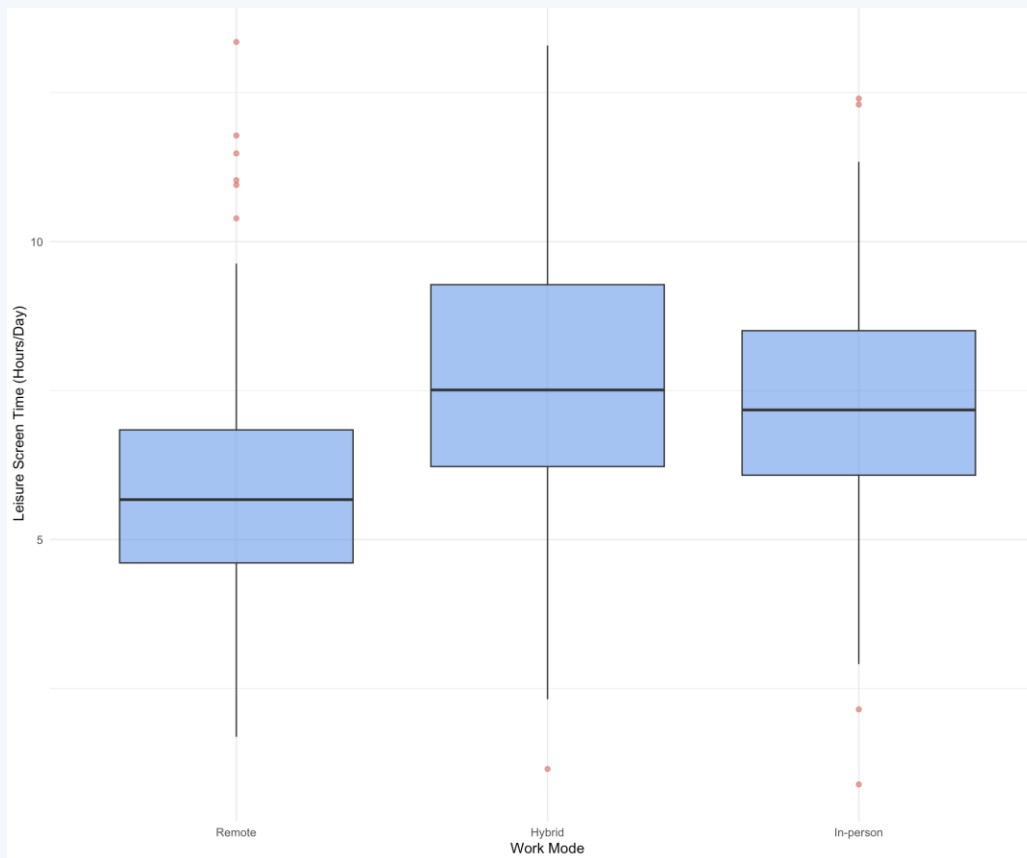
Leisure Dominates Screen Time Across All Occupations

Retired individuals have the largest share of work screen time



Does Work Mode Affect Leisure Screen Time?

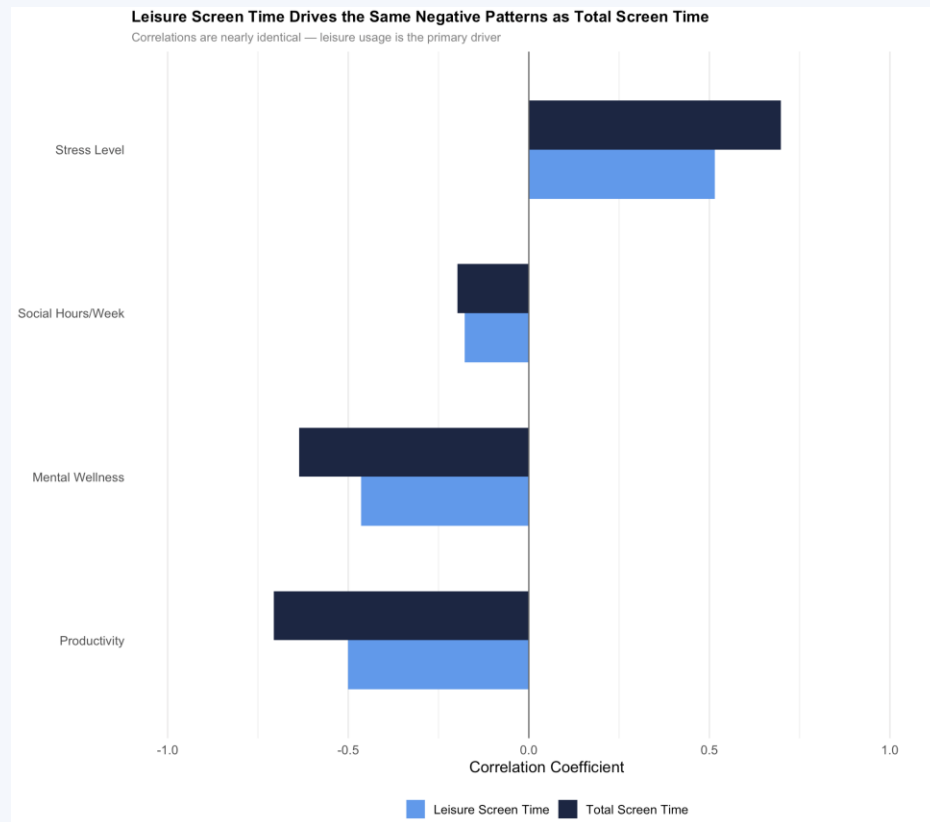
Key Finding: Hybrid and in-person workers have higher leisure screen time than remote workers, suggesting screens compensate for less screen-heavy work.



Leisure Screen Time — A Closer Look

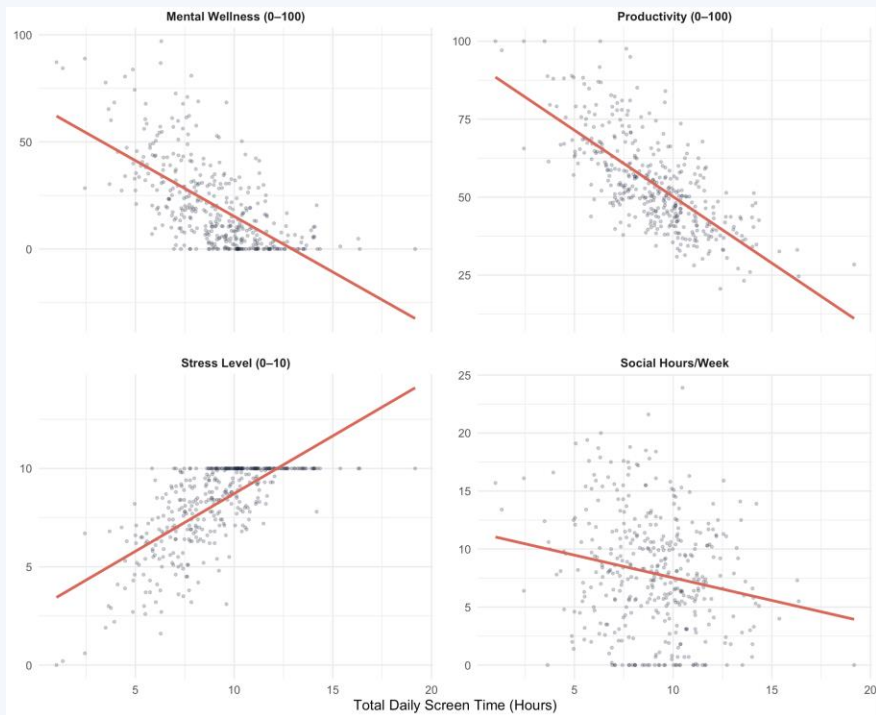
Key Finding: Leisure screen time shows the same negative trends as total screen time on mental wellness, productivity, and sleep. This suggests that leisure screen time is a key driver of harm.

→ Therefore, the argument that “I cannot lower my screen time because I need screens for work” is generally false because leisure screen time is as harmful as total screen time, and leisure screen time can be reduced.



Screen Time vs. Wellbeing Variables

Key Finding: More screen time consistently predicts lower mental wellness, lower productivity, less sleep, and higher stress — across all screen types.



Mental Wellness

Negative trend



Productivity

Negative trend



Stress Level

Positive trend

What Predicts Screen Time?

Stepwise Regression

Model Performance: $R^2 = 0.64$ — the model explains 64% of variance in screen time using 8 significant predictors.

Productivity

↓ reduces

Highest impact ($\beta = -0.52$)

Work Mode

↓ in-person least

Remote = more screen time

Stress Level

↑ increases

More stress → more screens

Occupation (Student)

↑ increases

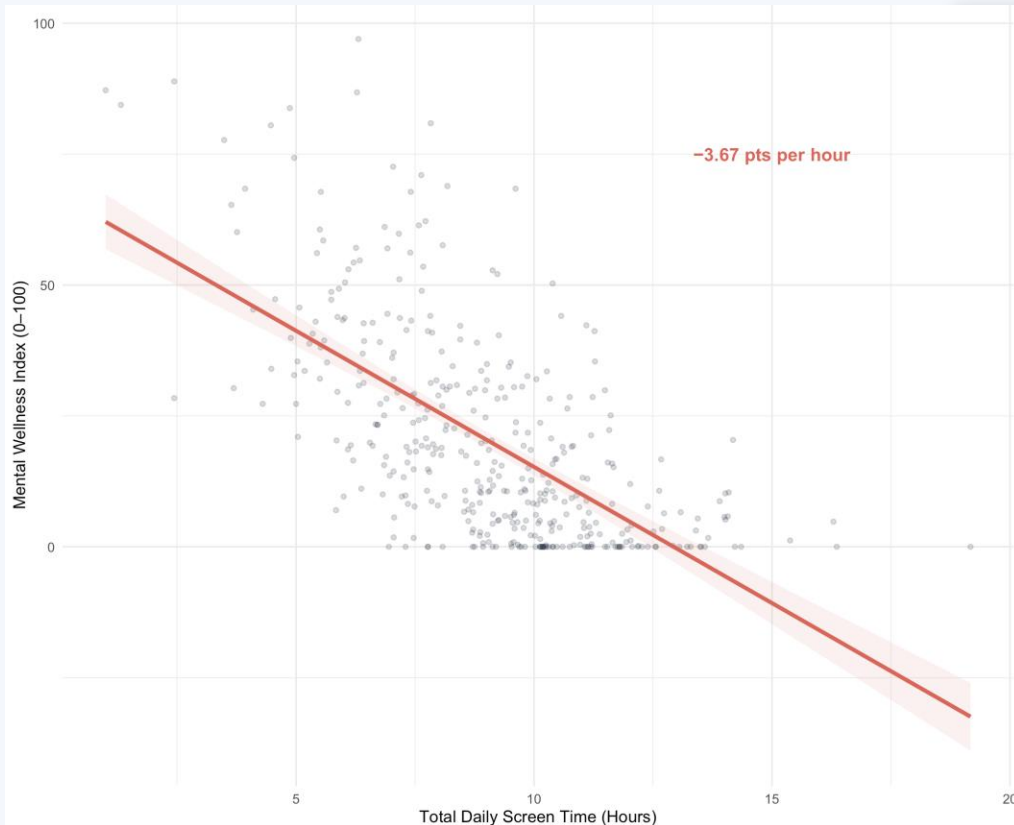
Strongest occupational effect

Social Hours

↓ reduces

Social activity buffers usage

Screen Time's Impact on Mental Wellness



-3.67

pts when controlling for social hours + interaction

-3.88

pts mental wellness per extra hour of screen time

Key Findings

01

Screen Time Harms Mental Wellness

Each additional hour of screen time is linked to a 3.88-point drop in mental wellness (0–100 scale), even after controlling for exercise and sleep quality.

02

Students & Remote Workers Are Most at Risk

Students accumulate the highest average screen time; remote workers follow. These groups experience the greatest exposure to screen time's negative effects, and any potential solutions would be best fit to these groups.

03

Social Activity Buffers the Harm

Adding social hours to the model shows that socializing partially offsets the negative impact of screen time, but does not fully eliminate it.

04

Stress & Productivity Also Affected

Screen time positively correlates with stress and negatively correlates with productivity, broadening the case for reducing screen usage.

What Should We Do About Screen Time?

- Reduce discretionary/leisure screen time where possible. The data shows a consistent dose-response relationship with wellness.
- Prioritize in-person social interaction, which defends against the mental toll of screen exposure.
- Target solutions towards high risk groups, like students

Limitations:

1. This dataset is observational; correlation does not imply causation.
2. All data is self-reported and not necessarily representative of national or broader trends